

SOLAR CELLS

An entropy-regulating molecular lock stabilizes formamidinium lead halide perovskite

Tianyin Miao^{1,2†}, Sanwan Liu^{3,4*}, Xia Lei^{4,5†}, Yong Zhang⁶, Wenpei Li¹, Qisen Zhou¹, Jianan Wang¹, Nikita A. Emelianov⁷, Victoria V. Ozerova⁷, Valeria S. Bolshakova⁷, Wenqiang Wang¹, Zheng Zhou¹, Zhongjie Zhu⁸, Lanlu Lu⁸, Zhenhua Chen⁹, Jingyuan Ma⁹, Erxiang Xu¹⁰, Luyao Wang¹¹, Yunfei Li¹¹, Zhengtian Tan¹, Shijie Zheng¹, Guilin Liu¹², Lianbo Guo¹, Jingbai Li^{4*}, Yang Shen¹⁰, Pavel A. Troshin^{13,14,7*}, Sergey M. Aldoshin⁷, Zonghao Liu^{1,2*}, Nam-Gyu Park^{3,15*}, Wei Chen^{1,2*}

A critical limitation of formamidinium lead iodide (FAPbI₃) perovskite solar cells (PSCs) lies in the intrinsic instability of the ionic-covalent Pb–I octahedral lattice, relative to the unfavorable hexagonal δ -phase under operating conditions. We report an entropy-regulating molecular-lock strategy using 1-pyridin-3-ylmethyl-piperazine hydrochloride (3-PMPCl). Strong interactions between the perovskite lattice surface and 3-PMPCl modulate the rotational freedom of organic cations and suppress the detrimental entropy increase associated with [PbI₆]^{4–} octahedra disorder or expansion. This entropy-favored environment intrinsically increases the phase transition energy barrier. The uniform distribution and strong adsorption of 3-PMPCl stabilize the α -phase under elevated temperature and humidity conditions. We achieved a certified power conversion efficiency (PCE) of 27.6% in FAPbI₃-based PSCs. However, the operational stability of such champion devices remains below the state of the art. Adopting a stable bismuth electrode addresses this issue with a slight reduction in efficiency, yielding a device that retains 93.0% of its initial PCE (26.8%) after 1011 hours at 85°C under 1-sun illumination.

The favorable bandgap and excellent chemical and thermal stability make formamidinium lead iodide (FAPbI₃) perovskite one of the most promising absorbers for high-efficiency perovskite solar cells (PSCs) (1–3). However, the photoactive α -phase can only achieve entropy stabilization through FA⁺ reorientation above 150°C (4), and the detrimental entropy increase accompanying the disorder and expansion of the inorganic [PbI₆]^{4–} octahedral framework also introduces a thermodynamically unfavorable contribution. The combined effect increases the Gibbs free energy of the cubic α -phase, making it metastable (4, 5). This thermodynamic instability facilitates a spontaneous transition that overcomes the phase transition energy barrier (E_b) to the asymmetric hexagonal nonphotoactive δ -phase (5–7). Mitigating this phase transition is imperative for achieving long operational lifetimes.

Lattice doping can enhance α -phase structural stability by reducing its Gibbs free energy. Indirect regulation of lattice interactions has

been closely linked to the biased displacement of [PbI₆]^{4–} octahedra, and secondary effects occur through induced slight lattice compression and disorder (6, 8–10). Nevertheless, compositional tuning often widens the bandgap and causes compositional inhomogeneity (5) that creates undoped FAPbI₃ regions that remain highly susceptible to δ -phase formation (11–13), potentially triggering a domino effect that results in the complete structure collapse (14, 15). To circumvent these limitations, we propose a molecular-lock stabilization strategy aimed at enhancing the intrinsic phase stability of unalloyed, pure-phase FAPbI₃ from kinetic and thermodynamic perspectives by increasing the favorable FA⁺ cation rotational entropy and suppressing the detrimental entropy increase associated with [PbI₆]^{4–} octahedra disorder or expansion while preserving its optoelectronic integrity.

Theoretical investigation on phase transition

The ultrasoft nature of FAPbI₃ is characterized by weak elastic interactions between FA⁺ cations and the [PbI₆]^{4–} octahedral framework. This readily facilitates Pb–I bond rupture and atom displacement, which contribute to an unfavorable structural entropy increase, while limiting the beneficial FA⁺ cation rotational entropy, thereby driving the α -to- δ phase transition (6, 7, 16, 17). To modulate lattice dynamics, suppressing large-amplitude thermal displacements of atoms is essential. We therefore focus on molecular additives featuring bifunctional groups and dimensions matched to the perovskite lattice. Such features could enhance molecular adsorption and lattice coherence, thereby strengthening the confinement of the lattice framework (16). Molecular anchoring or locking also requires conformational flexibility to allow optimization of the adsorption geometry under thermal lattice vibrations. On the basis of these considerations, we strategically selected a piperazine ring as one anchoring unit owing to its structural flexibility and demonstrated ability to interact with defective perovskite surfaces (18, 19). To further achieve lattice dimension matching and enhance molecular flexibility, we incorporated a methylene group into the molecular backbone. For the opposing terminal group, we introduced a pyridine group known for its strong Pb²⁺ coordination capability and systematically varied its nitrogen atom-position (2-/3-/4-N), with the benzene ring as reference (Fig. 1A) to probe position-dependent Pb²⁺ coordination effects on perovskite phase stability.

We first used density functional theory (DFT) calculations to assess the influence of these additives on the E_b in both the first lattice layer (referred to as the “surface”) and the second lattice layer (referred to as the “bulk”). To accurately model additive–lattice interactions, we used electrically neutral molecular counterparts of the four additives in our simulations (20). Pure α -FAPbI₃ must overcome energy barriers of 0.75 eV (surface) and 1.96 eV (bulk) to transition to the δ -FAPbI₃ phase (Fig. 1B and fig. S1). Weakly adsorbing *N*-benzylpiperazine hydrochloride (BPCl) and 1-pyridin-2-ylmethyl-piperazine hydrochloride (2-PMPCl) showed minimal enhancement in surface E_b , whereas strongly adsorbing 1-pyridin-3-ylmethyl-piperazine hydrochloride (3-PMPCl) and 1-pyridin-4-ylmethyl-piperazine hydrochloride (4-PMPCl) increased surface E_b up to sevenfold (Fig. 1C), indicating that the confinement effect of these molecules on the lattice surface depends on their adsorption strength (fig. S2). However, 4-PMPCl had a lower bulk E_b (2.75 eV) that we attributed to its inferior confinement effect on Pb atomic displacement, making the internal lattice prone to volumetric

¹Wuhan National Laboratory for Optoelectronics (WNLO) and School of Optical and Electronic Information, Huazhong University of Science and Technology (HUST), Wuhan, China. ²Optics Valley Laboratory, Hubei, China. ³School of Chemical Engineering and Center for Antibonding Regulated Crystals, Sungkyunkwan University (SKKU), Suwon, Republic of Korea. ⁴Hoffmann Institute of Advanced Materials, Shenzhen Polytechnic University, Shenzhen, China. ⁵Department of Materials Science and Engineering, Southern University of Science and Technology, Shenzhen, China. ⁶Sustainable Energy and Environment Thrust, Function Hub, The Hong Kong University of Science and Technology (Guangzhou), Guangzhou, China. ⁷Federal Research Center for Problems of Chemical Physics and Medicinal Chemistry of Russian Academy of Sciences, Academician Semenov Ave., Chernogolovka, Russia. ⁸National Facility for Protein Science in Shanghai, Shanghai Advanced Research Institute, Chinese Academy of Sciences, Shanghai, China. ⁹Shanghai Synchrotron Radiation Facility (SSRF), Shanghai Advanced Research Institute, Chinese Academy of Sciences, Shanghai, China. ¹⁰State Key Laboratory of New Ceramic Materials, School of Materials Science and Engineering, Tsinghua University, Beijing, China. ¹¹College of Chemistry and Chemical Engineering, Xiamen University, Xiamen, China. ¹²School of Science, Jiangnan University, Wuxi, China. ¹³Zhengzhou Advanced Research Institute of Harbin Institute of Technology (ZARI HIT), Zhengzhou, Henan, China. ¹⁴Harbin Institute of Technology (HIT), Harbin, China. ¹⁵SKKU Institute of Energy Science & Technology (SIEST), Sungkyunkwan University, Suwon, Republic of Korea. *Corresponding authors. Email: liusanwan@skku.edu (S.L.); lijingbai@szpu.edu.cn (J.L.); troshin@hit.edu.cn; troshin@icp.ac.ru (P.A.T.); liuzonghao@hust.edu.cn (Z.L.); npark@skku.edu (N.-G.P.); wnlochenwei@hust.edu.cn (W.C.) †These authors contributed equally to this work.

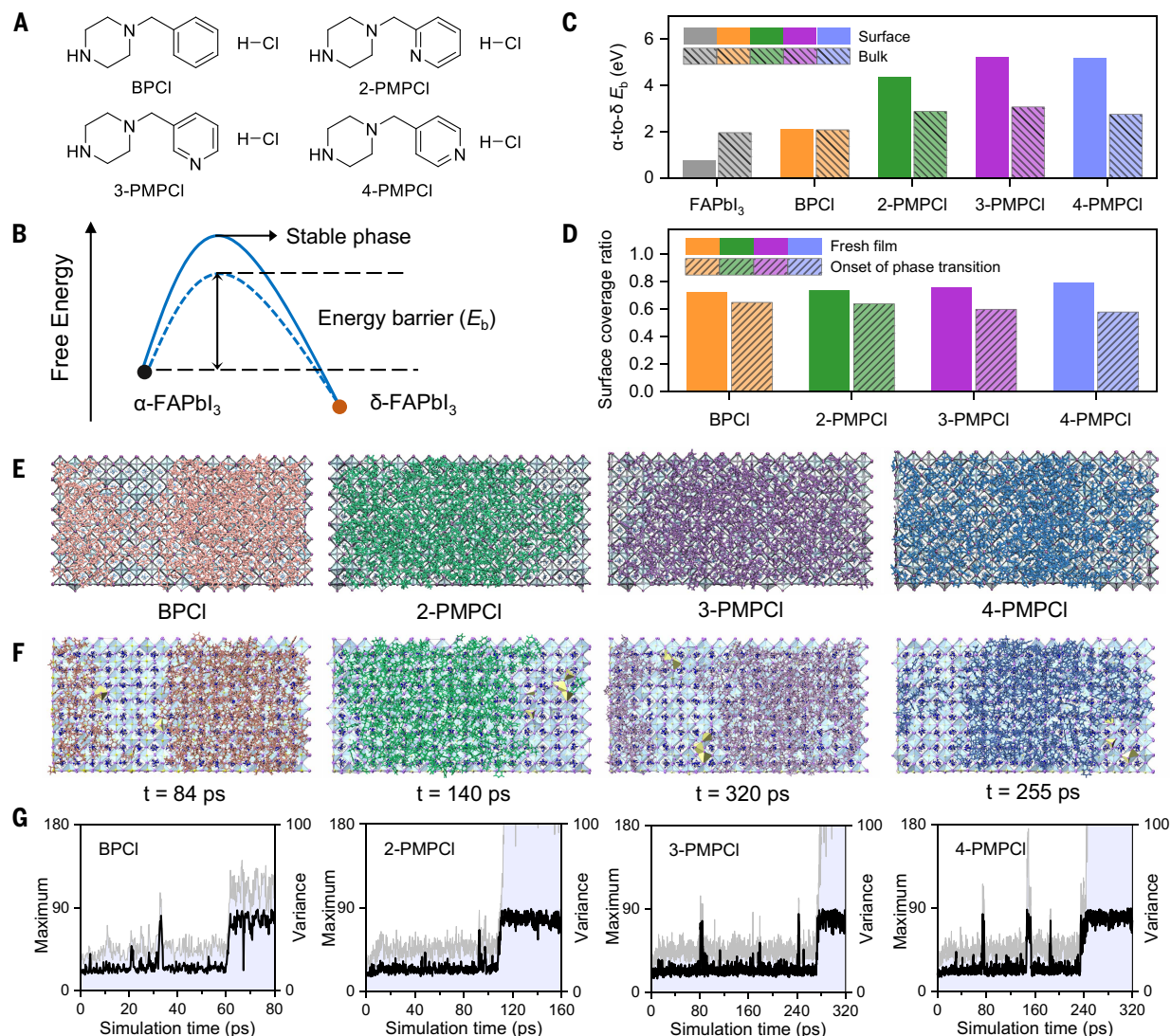


Fig. 1. Theoretical calculation of the α -to- δ phase transition of FAPbI₃. (A) Chemical structures of BPCI, 2-PMPCl, 3-PMPCl, and 4-PMPCl. (B) Schematic illustration of the Gibbs free energy relationship between α -FAPbI₃ and δ -FAPbI₃ at room temperature. The dashed line indicates that the metastable α -FAPbI₃ with higher Gibbs free energy tends to transform to δ -FAPbI₃ with lower Gibbs free energy by overcoming E_b . The solid line represents the energy-managed phase stabilization strategy achieved by increasing the energy barrier. (C) Surface and bulk E_b values of pure FAPbI₃ and FAPbI₃ modified with four different additives. (D) Surface molecular coverage of fresh films at 400 K and of films at the onset of phase transition at 573.15 K. (E) Model illustration of four different additives covering the surface of FAPbI₃. (F) Model illustration of four different additives covering the surface of FAPbI₃ at the onset of the phase transition. (G) Time required for the onset of the phase transition in FAPbI₃ modified with four different additives. The I–Pb–I bond angles on the perovskite surface were statistically analyzed every 0.2 ps. The maximum I–Pb–I bond angle at each time point was recorded, and the variance of bond angles was calculated. The phase transition was considered to begin when the I–Pb–I bond angles were consistently maintained at 90° for an extended period. The black lines represent the maximum bond angle, and the light grey lines represent the variance.

changes (figs. S3 and S4 and table S1) (21, 22). By contrast, as shown in fig. S4 and table S1, the superior dual-terminal N–Pb coordination of 3-PMPCl effectively confined lattice expansion, which increased the bulk E_b to 3.07 eV, surpassing that of BPCI (2.08 eV) and 2-PMPCl (2.88 eV) (Fig. 1C and figs. S2 and S3).

Given the prolonged dynamics of the phase transition process involving lattice dissociation and reconstruction (4, 5), uniform and persistent additive anchoring is crucial for α -phase stability. We further performed molecular dynamics (MD) simulations to assess the adsorption process of additives at 400 K. The surface coverages on the perovskite lattice of BPCI, 2-PMPCl, 3-PMPCl, and 4-PMPCl were 72, 74, 76, and 79%, respectively (Fig. 1, D and E). Such high coverages are beneficial for mitigating structural collapse triggered by local phase transitions (23, 24). Accelerated phase transition simulations further showed

coverage reduction to 65% (BPCI), 64% (2-PMPCl), 60% (3-PMPCl), and 58% (4-PMPCl) at the critical transition point (Fig. 1, D and F; δ -phase domains are marked in yellow in fig. S5). Thus, 3-PMPCl and 4-PMPCl not only exhibit strong dynamic adsorption capabilities but also stabilize the α -phase effectively, even at lower surface coverage.

During 500-ps accelerated phase-transition MD simulations, continuous lattice vibrations induced dynamic variations in the I–Pb–I bond angles. We systematically recorded the maximum I–Pb–I bond angle at each time point and calculated the corresponding variance of bond angles. The prolonged stabilization of bond angles at 90° (fig. S6) was identified as a signature of δ -phase formation. Because of the differences in intrinsic phase transition inhibition, dynamic adsorption, and stable functioning capability, the emergence time of the δ -phase varied greatly (fig. S7 and Fig. 1G). Pure FAPbI₃ underwent

phase transition at 48 ps, whereas BPCL, 2-PMPCl, and 4-PMPCl extended this transition to 62, 114, and 235 ps, respectively (Fig. 1G). Notably, 3-PMPCl substantially delayed δ -phase emergence to 277 ps with structural integrity well preserved after 510 ps, demonstrating superior phase-transition suppression (Fig. 1G and fig. S8).

Time-of-flight secondary ion mass spectrometry results confirmed the uniform vertical distribution of 3-PMPCl throughout the perovskite film, rather than confinement to specific depths (fig. S9). X-ray diffraction (XRD) analysis demonstrated that, below the additive concentration limit of 3-PMPCl in FAPbI₃ precursor solution (≤ 0.12 mol%), no low-dimensional phase formation occurred (fig. S10). Critically, 3-PMPCl-incorporated films exhibited no detectable δ phase after either ~ 860 hours of aging in N₂ at room temperature or 200 hours of thermal aging at 85°C [$\sim 40\%$ relative humidity (RH)] and remained unchanged even after 40 days of aging in ambient air. Such phase stability notably surpasses that achieved with the other three additives or representative reported efficiency and stability-enhancing molecules (figs. S11 and S12).

Entropy-favored octahedral lattice

We conducted comprehensive characterization of perovskite films to elucidate the 3-PMPCl-induced phase transition suppression mechanism. Fresh 3-PMPCl-incorporated films exhibited a pure α -phase crystalline structure identical to that of pristine films (fig. S13), with a negligible influence on bandgap and morphology (figs. S14 and S15). However, x-ray photoelectron spectroscopy revealed a distinct shift in Pb 4f and I 3d peaks toward lower binding energies (fig. S16), accompanied by a transition of internal stress from tensile stress to slight compressive stress verified by grazing incidence XRD characterization (fig. S17), confirming 3-PMPCl interaction with uncoordinated Pb²⁺ (9, 13, 25).

Solution-phase nuclear magnetic resonance (NMR) spectroscopic analysis further provided evidence for the interactions of perovskite composition with 3-PMPCl. First, a strong association with FA⁺ from FAI was observed, and a hydrogen-bonded complex was formed with a fixed geometry that could contribute to a long-term stability (fig. S18). This complexation suppresses N–H proton exchange, creating diastereotopic protons that manifest as severely split FA⁺ signals in the ¹H NMR spectrum. Second, the chelation of Pb²⁺ cations from PbI₂ was indicated by notable chemical-shift perturbations of the characteristic resonances of 3-PMPCl (fig. S19). Infrared scattering-type scanning near-field optical microscopy (IR s-SNOM) enabled direct nanoscale mapping of the distribution of key organic species in the perovskite films (26, 27). Our analysis revealed that the 3-PMPCl additive was predominantly localized at perovskite grain boundaries (Fig. 2, A and B, and fig. S20), facilitating surface stabilization and passivation of defects such as uncoordinated Pb²⁺ cations.

The ²H solid-state NMR peak shape exhibits high sensitivity to cation dynamics (28). The 3-PMPCl-modified sample exhibited further narrowing of the ²H spectral peak (Fig. 2C), indicating that FA⁺ cations undergo faster and less constrained reorientational motion on 3-PMPCl locking (28). Furthermore, we used polarized Raman spectroscopy to investigate changes in the orientational configurations of FA⁺ cations. In the control sample, FA⁺ vibrational signals at 1384 cm^{−1} (assigned to NH₂⁺ rocking, ρ) and 1117 cm^{−1} (assigned to C–N symmetric stretching, ν_s) became nearly undetectable at specific angles (fig. S21), indicating that FA⁺ vibrational coupling requires specific alignment with the incident light polarization (29, 30). By contrast, the angular dependence was suppressed in the 3-PMPCl-modified sample, reflecting enhanced dynamic disorder. This enhancement in cation rotational entropy was quantitatively confirmed by MD simulations. By tracking the orientational distribution of FA⁺ cations across the eight octants of a three-dimensional Cartesian coordinate system during the pre-transition trajectories, we calculated the mean rotational entropy per FA⁺ via the Boltzmann entropy formula. As shown in Fig. 2D and fig. S22,

the modification of 3-PMPCl led to higher FA⁺ cation rotational entropy that persisted for a long time, consistent with our experimental results.

To further investigate the impact of the modified chemical environment on lattice stability, we conducted extended x-ray absorption fine structure (EXAFS) measurements for tracking the structural changes in the corresponding films before and after 15 days of aging under pure nitrogen conditions at room temperature. Synchrotron x-ray was incident on perovskite films at 1° with data collected in fluorescence mode. As shown in fig. S23A, Pb L_{III}-edge absorption energies of control and 3-PMPCl-incorporated films remained unchanged before and after aging, confirming the persistent Pb²⁺ oxidation state (31). Furthermore, the x-ray absorption near-edge structure spectra maintained consistent intensities, indicating that the primary composition of the perovskite phase remained stable throughout the aging process (13, 31).

To precisely assess the structural impacts of 3-PMPCl, we conducted detailed EXAFS analysis involving k^2/k^3 -weighted fitting (where k is the wave vector) and Fourier transformations of k -space EXAFS data to generate the real (R)-space curves (fig. S23, B to K, and table S2). We performed fitting using the model with the lowest misfit factor, as shown in fig. S23, B and G, and table S2. A relatively weak peak around 2.2 Å corresponds to the Pb–O bond, and a strong peak around 2.8 Å corresponds to the Pb–I bond (13, 31–33). The derived Pb–I bond distances were 3.144 Å for fresh control films and 3.143 Å for 3-PMPCl-modified films, demonstrating that the incorporation of 3-PMPCl had no adverse impact on the short-range order of the perovskite lattice (13).

After ~ 400 hours of aging under nitrogen at room temperature, the Pb–I bond distances shifted to 3.162 and 3.149 Å for control and 3-PMPCl-modified films, respectively (Fig. 2E). This contrast demonstrated progressive Pb–I bond strength weakening in the control film during aging, whereas 3-PMPCl incorporation effectively suppressed detrimental Pb–I bond elongation and distortion, thus inhibiting thermodynamically driven structural expansion and reorganization of the framework. Further evidence of stabilization was obtained from Debye–Waller (D–W) factor analysis (Fig. 2F). Fresh films showed comparable D–W factors (0.0191 for control and 0.0190 for the 3-PMPCl case), indicating a similar initial lattice distortion level (34). However, the D–W factor for the control film increased to 0.0213 after aging, whereas the 3-PMPCl film remained nearly changed, providing definitive evidence of its structural stabilization capability (Fig. 2F). The improved structural stability led to an increase in ion migration activation energy (fig. S24), thereby effectively enhancing phase stability, in good agreement with our simulation results (35).

To further elucidate the phase stabilization mechanism of 3-PMPCl, we conducted in situ grazing-incidence wide-angle x-ray scattering (GIWAXS) measurements of perovskite films under damp-heat conditions (85°C and 85% RH). The evolution of characteristic diffraction peaks along the out-of-plane [$\mathbf{q}_z$, corresponding to the (001) plane] and in-plane [$\mathbf{q}_{xy}$, corresponding to the (010) plane] scattering vectors (~ 1 Å^{−1}) differed between the control film and 3-PMPCl-modified film (Fig. 2, G and H). The control film exhibited progressive shifts of both $\mathbf{q}_z$ and $\mathbf{q}_{xy}$ to lower values, indicating the expansion of (001) and (010) interplanar spacings and consequent [PbI₆]^{4−} octahedra dilation (fig. S25) (22, 36). The observed reduction in symmetry and the widening of the characteristic diffraction peak indicated an increase in octahedral distortion, which is typically associated with the formation of the δ -phase (22). By contrast, the 3-PMPCl-incorporated film maintained constant peak positions and widths throughout testing, demonstrating effective “locking” of the [PbI₆]^{4−} octahedra framework against moisture- and thermally induced deformation. After 200 min of damp-heat testing, the control film showed a slight δ -phase signature (Fig. 2I), whereas the 3-PMPCl sample showed no detectable δ -FAPbI₃ signal (Fig. 2J). Notably, the 3-PMPCl-modified film demonstrated a superior out-of-plane crystallographic orientation (fig. S26), which has been confirmed to facilitate enhanced charge carrier transport (37).

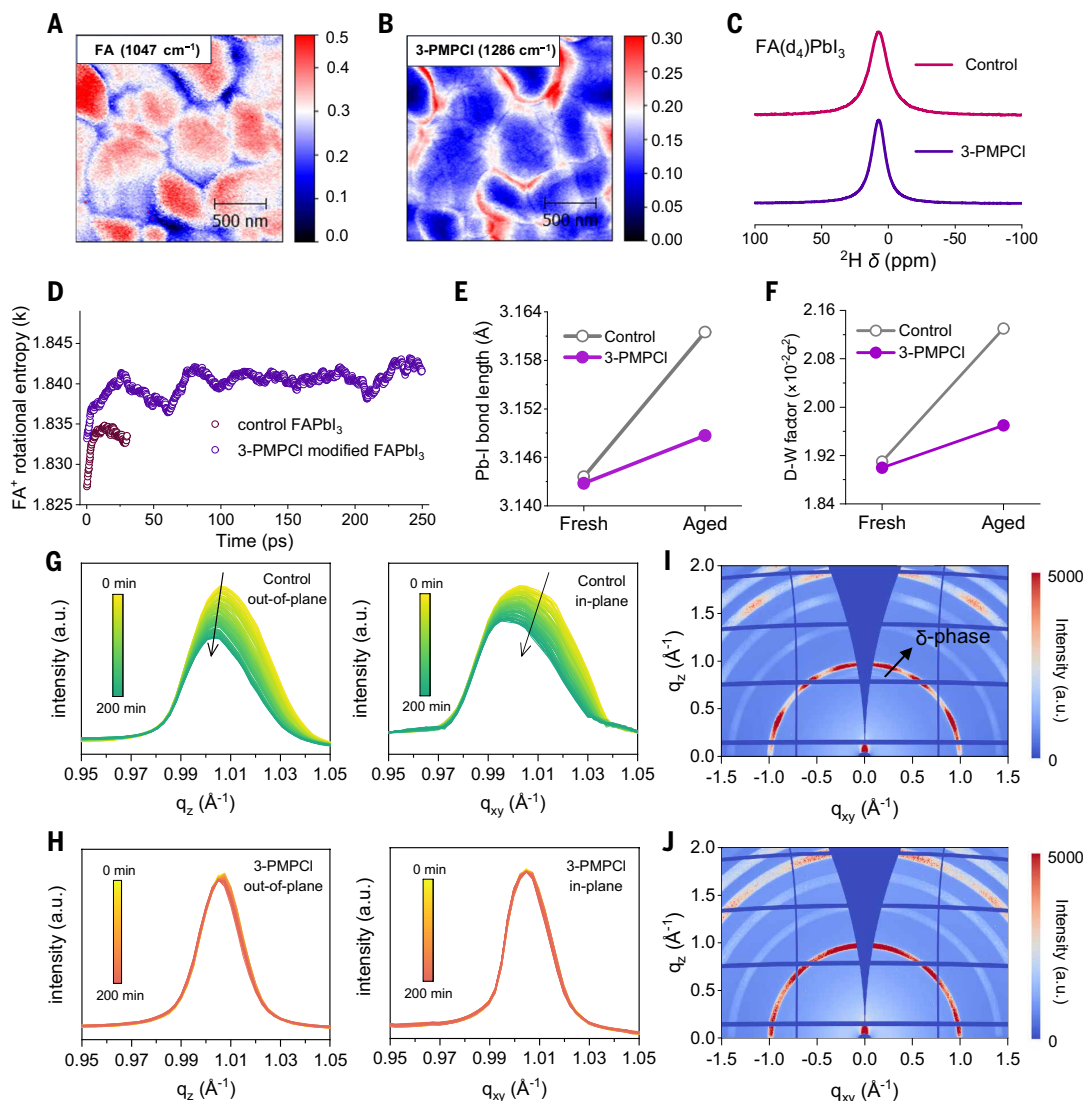


Fig. 2. Entropically favored octahedral lattice mechanism. IR s-SNOM images (red and blue color codes) showing the spatial distribution of (A) FA⁺ cations and (B) 3-PMPCl additive. (C) ²H solid-state NMR spectra of control and 3-PMPCl-modified deuterated FAPbI₃. (D) Quantification of FA⁺ dynamic disorder via rotational entropy, derived from the Boltzmann-weighted distribution of its orientations in Cartesian space before phase transition. (E) Pb-I bond length and (F) D-W factor of control and 3-PMPCl-modified samples from the fit result of the EXAFS spectra. Integrated out-of-plane and in-plane profiles obtained from in situ GIWAXS maps for (G) the control film and (H) the 3-PMPCl-modified film aged at 85°C and ~85% RH for 200 min. Two-dimensional GIWAXS patterns of (I) the control film and (J) 3-PMPCl-modified film after aging.

Complementary Fourier-transform infrared analysis of samples under thermal aging (85°C, 2 hours) in ambient air provided molecular-level insights (fig. S27). The control film exhibited pronounced changes in the intensity and broadness of the characteristic N-H and C-H vibrational peaks after aging. By contrast, the 3-PMPCl-incorporated film displayed negligible variation, indicating that the release and orientational rearrangements of FA⁺ cations were effectively suppressed (10). The long-term anchoring characteristic of 3-PMPCl for perovskite stability improvement was further verified by the ultraviolet-visible absorption evolution (fig. S28).

Optoelectronic properties of perovskite films

We systematically investigated the passivation mechanism of four molecular additives using DFT calculations. By modeling four characteristics of perovskite surface defects—I-vacancy (V_I), Pb-vacancy (V_{Pb}), and Pb-I antisites (Pb_I and I_{Pb})—we quantified their interactions with each molecule (figs. S29 to S31 and Fig. 3A). Notably, 3-PMPCl exhibited the strongest Pb²⁺ passivation by N_{pyridine}, thus promoting V_I passivation,

the primary defects driving phase transitions, which also provides a molecular-level explanation for its exceptional phase stabilization capability (38). In addition, 3-PMPCl also exhibited higher interaction energies with other defects than BPCl and 2-PMPCl and maintained performance comparable to that of 4-PMPCl.

Optical characterization confirmed the above computational predictions: The 3-PMPCl-incorporated films exhibited enhanced steady-state photoluminescence (PL) intensity (Fig. 3B), longer carrier lifetime in time-resolved PL measurements (Fig. 3C and table S3), and higher carrier mobility in time-resolved microwave conductivity measurements (fig. S32). These enhancements were attributed to the formation of the entropy-favored lattice, which improves structural order and suppresses large fluctuations in bond lengths and atomic displacements to intrinsically reduce defects and enhance charge transport, together with the effective defect-passivation effect of 3-PMPCl, collectively reducing defect density (fig. S33) and effectively suppressing defect-induced nonradiative recombination. Spatial imaging through PL mapping and laser confocal fluorescence lifetime imaging microscopy

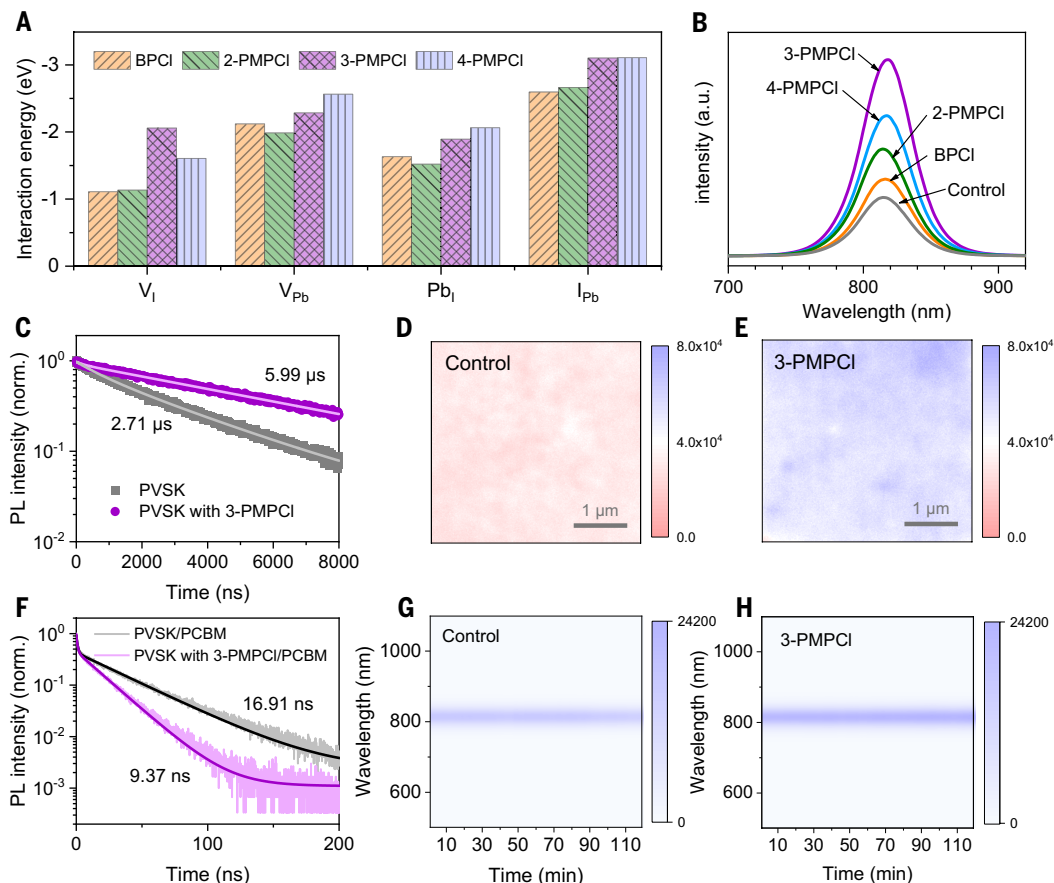


Fig. 3. Enhancement of perovskite film quality. (A) The interaction energy between BPCI, 2-PMPCl, 3-PMPCl, and 4-PMPCl and four typical defects, including I-vacancy (V_I), Pb-vacancy (V_{Pb}), and Pb–I antisites (Pb_I and I_{Pb}). (B) Steady-state PL spectra of the control, BPCI-, 2-PMPCl-, 3-PMPCl-, and 4-PMPCl-modified films. (C) Time-resolved PL spectra of the control film and 3-PMPCl-modified film. Solid lines denote the fitted data curves. (D and E) PL mapping images of the control film (D) and the 3-PMPCl-modified film (E). (F) Time-resolved PL spectra of the glass/perovskite (PVSK)/PCBM samples with and without 3-PMPCl incorporation. Solid lines denote the fitted data curves. (G and H) In situ PL spectra of the control film (G) and the 3-PMPCl-modified film (H) under continuous laser irradiation in ambient air for 120 min at 85°C and ~85% RH.

further revealed improved uniformity in PL intensity and carrier lifetime in 3-PMPCl-modified films (Fig. 3, D and E, and fig. S34). These results indicate that a uniformly low defect density was achieved in the 3-PMPCl-modified film, which mitigated phase transitions from defect-rich regions (39, 40). To further quantify the reduction in non-radiative recombination losses, we measured the photoluminescence quantum yield (PLQY) of film samples (fig. S35). The PLQY increased from 1.12% (control film) to 6.92% (3-PMPCl film), indicating a 47.1-meV enhancement in quasi-Fermi level splitting ($\Delta QFLS$) (3-PMPCl versus control). This improvement in $\Delta QFLS$ would be expected to improve the open-circuit voltage (V_{OC}) in PSCs.

The faster photocurrent decay observed in the 3-PMPCl-incorporated device, together with the accelerated PL quenching in the 3-PMPCl-incorporated film on contact with [6,6]-phenyl-C61-butyric acid methyl ester (PCBM), indicates improved charge extraction at the perovskite/PCBM interface (Fig. 3F, fig. S36, and table S4). Ultraviolet photoelectron spectroscopy measurements revealed the influence of 3-PMPCl on the perovskite surface electronic structure (fig. S37 and table S5). The incorporation of 3-PMPCl induced a slight shift in the Fermi level (E_F) toward the conduction band minimum, enhancing n-type characteristics. This modification, likely arising from altered surface termination, optimized the energy level alignment at the perovskite/PCBM interface, contributing to a slight boost in charge extraction and device performance (41).

The stability of these interfacial improvements was rigorously evaluated through in situ PL measurements under harsh environmental

conditions (85°C, ~85% RH). After 120 min of stress testing, 3-PMPCl-modified films maintained high PL intensity with minimal variation, in stark contrast to control samples which exhibited severe PL quenching (Fig. 3, G and H, and fig. S38). Meanwhile, the film maintained uniform and high PL intensity after 15 days of aging in an N_2 atmosphere (fig. S39). These results demonstrate that 3-PMPCl not only improves charge extraction but also maintains this advantage under operational stresses. To provide microscopic visualization of the stability enhancement, we conducted in situ IR s-SNOM characterization to directly probe the photostability of perovskite films. Perovskite samples with and without 3-PMPCl were subjected to blue light (460 nm, ~100 mW cm⁻²) on the microscope stage, enabling the direct observation of organic cation dynamics during aging. The analysis revealed rapid photodegradation in control films, evidenced by the fast depletion of FA⁺ cations. Conversely, films modified with 3-PMPCl showed markedly increased stability, with no substantial loss of FA⁺ cations from the surface (movie S1). The combination of enhanced interfacial charge transport and exceptional interfacial stability positions 3-PMPCl as an effective molecular modifier for high-performance, durable perovskite photovoltaics.

Device photovoltaic and stability performance

We fabricated inverted devices with the architecture of glass/indium tin oxide/nickel oxide (NiO_x)/self-assembled monolayers/FAPbI₃/PCBM/SnO₂/Bi, selected for its exceptional interfacial stability to ensure reliable long-term assessment (25, 42). Detailed fabrication procedures

are comprehensively outlined in the supplementary materials. The champion 3-PMPCl-incorporated FAPbI₃-based PSC achieved a power conversion efficiency (PCE) of 26.8%, with a V_{OC} of 1.20 V, a short-circuit current density (J_{SC}) of 26.1 mA cm⁻², and a fill factor (FF) of 85.7% (device area/measured mask area: 0.0745 cm²/0.0556 cm²), outperforming the control device (PCE = 26.0%, V_{OC} = 1.18 V, J_{SC} = 26.0 mA cm⁻², and FF = 84.7%) (Fig. 4A). The average PCE of 3-PMPCl-incorporated devices reached 26.6%, with statistical analysis confirming that the enhancement arises from gains in both V_{OC} and FF (fig. S40).

To further probe the efficiency potential of 3-PMPCl, we also fabricated devices using the conventional BCP/Ag buffer layer/electrode architecture (movie S2). Benefiting from enhanced electrode conductivity and higher optical reflectivity, the device performance was further improved (Fig. 4A). The 3-PMPCl-incorporated FA-based and formamidinium-cesium (FACs)-based devices achieved efficiencies of 27.6% (certified 27.6%) and 27.5% (certified 27.4%), respectively (Fig. 4, B to E). A detailed analysis of the photovoltaic characterization is presented in

supplementary text, figs. S41 to S51, and tables S6 to S9. These values represent the highest certified efficiencies reported to date for both FA-based and FACs-based PSCs (Fig. 4, D and E, and tables S7 and S9), with excellent reproducibility (fig. S51).

The product $V_{OC} \times FF$ relative to the corresponding Shockley–Queisser (S–Q) limit allows for the quantification of the trade-off between V_{OC} and FF without the influence of different bandgaps (43, 44). Our devices achieved certified $V_{OC} \times FF$ values of 91.8% (FAPbI₃) and 90.9% (FACsPbI₃) of their respective S–Q limit, representing the highest reported values for PSCs (Fig. 4F and table S10). These $V_{OC} \times FF$ values demonstrate that 3-PMPCl incorporation facilitated charge transport and suppressed nonradiative recombination, minimizing V_{OC} and FF losses to achieve the highest certified steady-state PCEs among all single-junction PSCs (table S11). Moreover, the V_{OC} –light intensity dependence (fig. S52) revealed a reduction in the ideality factor from 1.41 (control) to 1.20 by 3-PMPCl incorporation, quantitatively confirming the reduction of nonradiative recombination. The incorporation of

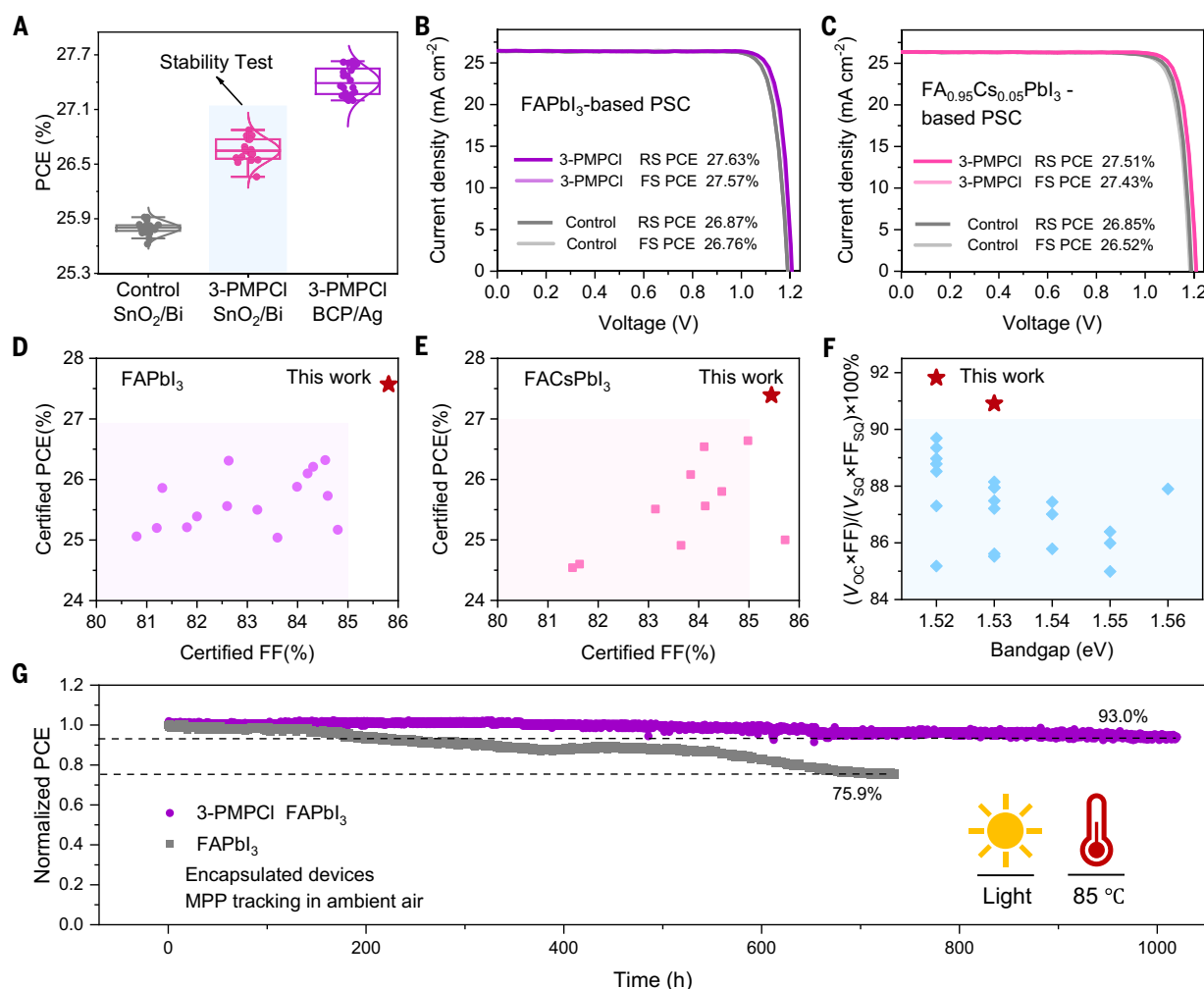


Fig. 4. PCEs and stability of the 3-PMPCl-modified PSCs. (A) PCE distribution of FAPbI₃-based PSCs. Data represent 20 devices each for the SnO₂/Bi architecture (without and with 3-PMPCl) and the BCP/Ag architecture with 3-PMPCl. (B) J - V curves of the champion FAPbI₃-based PSCs with and without 3-PMPCl incorporation measured at reverse scan (RS) and forward scan (FS). (C) J - V curves of the champion FA_{0.95}Cs_{0.05}PbI₃-based PSCs with and without 3-PMPCl incorporation measured at RS and FS. (D) Comparison of certified PCE and FF of our FAPbI₃-based PSCs with other reported high-performance FAPbI₃-based PSCs. (E) Comparison of certified PCE and FF of our FA_{0.95}Cs_{0.05}PbI₃-based PSCs with other reported high-performance FA_{0.95}Cs_{0.05}PbI₃-based PSCs. (F) Comparison of our certified $V_{OC} \times FF$ values, normalized to the Shockley–Queisser limit, with those of other reported high-performance PSCs as a function of bandgap energy. (G) Operational stability of encapsulated devices under continuous 1-sun-equivalent white-light LED illumination with MPPT in ambient air at 85 °C. The initial PCEs of the 3-PMPCl-modified FAPbI₃-based PSC and pure FAPbI₃-based PSC used for the operational stability tests were 26.8% and 25.9%, respectively.

3-PMPCl improved the built-in voltage (V_{bi}) from 1.07 to 1.11 V, further confirming enhanced charge separation and collection (fig. S53) (45).

After confirming the favorable contribution of 3-PMPCl incorporation to device optoelectronic performance, we measured the stability on the stable SnO_2/Bi architecture devices. Under damp-heat testing (85°C and 85% RH) following the International Electrotechnical Commission 61215:2016 standard, encapsulated 3-PMPCl devices maintained 95.7% of their initial PCEs (26.8%; fig. S54A) over 1150 hours of aging, whereas the control devices retained only 85.2% of their initial PCEs (26.0%; fig. S54B) over the same aging time (fig. S55). Moreover, we also examined the operational stability of the encapsulated devices with maximum power point tracking (MPPT) under continuous 1-sun-equivalent white-light light-emitting diode (LED) illumination at 85°C in ambient air (40 to 50% RH) (fig. S56). The 3-PMPCl-incorporated device retained 93.0% of its initial PCE (26.8%; fig. S54C) over 1019 hours of aging, whereas the control device retained only 75.9% of its initial PCE (25.9%; fig. S54D) after 734 hours of aging (Fig. 4G). One BCP/Ag-based device with an efficiency of 27.6% was also subjected to the same stability test. As we have previously reported (42), such architectures exhibit inherently poor stability, which persisted even after the incorporation of 3-PMPCl (fig. S57). These stability tests conclusively demonstrate that 3-PMPCl integration provides exceptional protection against thermally and photo-induced degradation pathways.

Discussion

Our investigation unveils that constructing a long-term stable entropy-favored lattice can intrinsically stabilize the photoactive α -phase from kinetic and thermodynamic perspectives. This is achieved by introducing suitable lattice constraints through dual-site dynamic flexible anchoring on the surface. This approach enhances the rotational entropy of FA^+ cations while suppressing the detrimental entropy increase associated with $[\text{PbI}_6]^{4-}$ octahedra disorder, thereby substantially delaying the detrimental phase transition. Furthermore, the entropy-favored lattice enhances structural order, which lowers defect density, reduces non-radiative recombination, and improves charge transport, ultimately leading to a certified champion efficiency of 27.6% in FAPbI_3 -based solar cells. However, the operational stability of such champion devices remains below the state of the art. Implementing a bismuth electrode architecture overcomes this issue with a slight loss of efficiency, yielding a device that retains 93.0% of its initial PCE (26.8%) after 1011 hours at 85°C under 1-sun illumination. The damp-heat stability is also enhanced.

REFERENCES AND NOTES

- H. Zai *et al.*, *Science* **387**, 186–192 (2025).
- S. Sidhik *et al.*, *Science* **384**, 1227–1235 (2024).
- S. Yu *et al.*, *Science* **382**, 1399–1404 (2023).
- T. Chen *et al.*, *Sci. Adv.* **2**, e1601650 (2016).
- T. A. S. Doherty *et al.*, *Science* **374**, 1598–1605 (2021).
- X. Liu *et al.*, *Nat. Rev. Chem.* **7**, 462–479 (2023).
- H.-S. Kim, N.-G. Park, *Adv. Energy Mater.* **15**, 2400089 (2025).
- S. Li *et al.*, *Nature* **635**, 82–88 (2024).
- L. Zheng *et al.*, *Science* **388**, 88–95 (2025).
- E. Ugur *et al.*, *Science* **385**, 533–538 (2024).
- S.-H. Turren-Cruz, A. Hagfeldt, M. Saliba, *Science* **362**, 449–453 (2018).
- Z. Liang *et al.*, *Nature* **624**, 557–563 (2023).
- Y. Ding *et al.*, *Science* **386**, 531–538 (2024).
- Z. Huang *et al.*, *Nature* **623**, 531–538 (2024).
- Z.-W. Gao *et al.*, *Nat. Photonics* **19**, 258–263 (2025).
- Y. Zou *et al.*, *Science* **385**, 161–167 (2024).
- Y. Zhang *et al.*, *Science* **387**, 284–290 (2025).
- D. B. Khadka *et al.*, *Nat. Commun.* **15**, 882 (2024).
- S. Mariotti *et al.*, *Science* **381**, 63–69 (2023).
- G. Kresse, J. Furthmüller, *Comput. Mater. Sci.* **6**, 15–50 (1996).
- C. Zhu *et al.*, *Nat. Commun.* **10**, 815 (2019).
- Y. Shen *et al.*, *Nature* **635**, 882–889 (2024).
- P. Shi *et al.*, *Nat. Commun.* **15**, 9703 (2024).
- J. Lim *et al.*, *Energy Environ. Sci.* **17**, 8209–8218 (2024).
- S. Liu *et al.*, *Nature* **632**, 536–542 (2024).
- S. Luo *et al.*, *Energy Environ. Sci.* **18**, 2436–2451 (2025).
- S. Cao *et al.*, *Adv. Energy Mater.* **15**, 2405367 (2025).
- A. Mishra, M. A. Hope, M. Grätzel, L. Emsley, *J. Am. Chem. Soc.* **145**, 978–990 (2023).
- X. Chen *et al.*, *Nat. Commun.* **16**, 9717 (2025).
- Z. Liang *et al.*, *Nat. Mater.* **25**, 267–274 (2026).
- C. Ma *et al.*, *Science* **379**, 173–178 (2023).
- Y. Wen *et al.*, *Nat. Commun.* **15**, 7085 (2024).
- D.-H. Kang, S.-U. Lee, N.-G. Park, *ACS Energy Lett.* **8**, 2122–2129 (2023).
- D.-H. Kang, Y.-J. Park, Y.-S. Jeon, N.-G. Park, *J. Energy Chem.* **67**, 549–554 (2022).
- J. Wang *et al.*, *Nat. Commun.* **16**, 4917 (2025).
- Z.-W. Gao *et al.*, *Joule* **8**, 255–266 (2024).
- Q. Zhou *et al.*, *Nat. Energy* **10**, 1371–1381 (2025).
- B. Wen *et al.*, *Nat. Commun.* **15**, 7720 (2024).
- X. Chu *et al.*, *Nat. Energy* **8**, 372–380 (2023).
- Z. Ni *et al.*, *Science* **367**, 1352–1358 (2020).
- S. Liu *et al.*, *Adv. Mater.* **37**, e2415100 (2025).
- J. Zhou *et al.*, *Nat. Commun.* **14**, 6120 (2023).
- S. Zhang *et al.*, *Science* **380**, 404–409 (2023).
- W. Peng *et al.*, *Science* **379**, 683–690 (2023).
- E. H. Jung *et al.*, *Nature* **567**, 511–515 (2019).
- T. M. Miao, An entropy-regulating molecular lock stabilizes formamidinium lead halide perovskite, version V1, Demo Dataverse (2026); <https://doi.org/10.70122/FK2/QLWOFH>.

ACKNOWLEDGMENTS

We thank the Analytical and Testing Center of HUST for the support of facilities for sample measurements. We thank the Center for Computational Science and Engineering at Southern University of Science and Technology and the Hoffmann Institute of Advanced Materials at Shenzhen Polytechnic University for providing the computing resources. We also thank the National Center for Magnetic Resonance in Wuhan, State Key Laboratory of Magnetic Resonance Spectroscopy and Imaging, We thank 14B, 20U2 RsoXS, and Hard Branch of BL20U beamline station at SSRF for providing time for x-ray diffraction measurements. We thank the staff of the BL17B beamline at the National Facility for Protein Science in Shanghai, Shanghai Advanced Research Institute, Chinese Academy of Sciences, for their technical support in GIWAXS data collection and analysis. We would like to thank Ezhou Industrial Technology Research Institute, HUST, for the facility support of PSC characterization. The contributions of N. A. Slesarenko and S. G. Vasilyev from the Federal Research Center for Problems of Chemical Physics and Medicinal Chemistry of the Russian Academy of Sciences for the NMR analysis are gratefully acknowledged. We thank the Harvard Dataverse for providing a reliable platform for sharing and archiving research data (46). **Funding:** This work was supported by the National Natural Science Foundation of China (W2412077 and 52473301), the Ministry of Science and Technology of China (2021YFB3800104), the Innovation Project of Optics Valley Laboratory (OVL2025YZ004 and OVL2021BG008), the National Science Foundation of Hubei Province (2022CFA093), the Fundamental Research Support Program of Huazhong University of Science and Technology (2025BRB016), the State Key Laboratory of Alternate Electrical Power System with Renewable Energy Sources (LAPS25001), the National Research Foundation of Korea (NRF) grants funded by the Korean government (MSIT) under contract NRF-2021R1A3B1076723 (Research Leader Program) and RS-2025-02316700 (Key Technology Development Project for Carbon-Free Energy), and Russian Science Foundation (project no. 19-73-30020P). **Author contributions:** T.M., S.L., and X.L. contributed equally to this work. W.C., N.G.P., Z.L., J.L., P.A.T., S.L., S.M.A., Y.S., and T.M. conceived the project and designed the experiments. W.C., N.G.P., and Z.L. directed and supervised the project. T.M. fabricated the devices. T.M., S.L., Y.Z., W.L., Q.Z., J.W., W.W., L.G., Z. Zhou., E.X., L.W., Y.L., Z.T., S.Z., and G.L. performed the material and device characterization. N.A.E. performed IR s-SNOM measurements. V.V.O. and V.S.B. contributed to the material synthesis and sample preparation for s-SNOM and NMR measurements. Z.C. and J.M. performed the EXAFS measurements. T.M., Z. Zhu., and L.L. performed the in situ GIWAXS measurements. S.L. and N.G.P. performed the temperature-dependent dark conductivity measurements. X.L. and J.L. performed the DFT calculations, MD simulations, and theoretical analysis. W.C., N.G.P., Z.L., J.L., S.L., and T.M. co-wrote the paper. All authors discussed the results and commented on the written manuscript. **Competing interests:** The authors declare that they have no competing interests or patents related to this work. **Data, code, and materials availability:** All data needed to evaluate the conclusions in this study and details of materials synthesis are present in the main text or the supplementary materials. **License information:** Copyright © 2026 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. <https://www.sciencemag.org/about/science-licenses-journal-article-reuse>

SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.aeb9953
Materials and Methods; Figs. S1 to S57; Tables S1 to S11; References (47–57); Movies S1 and S2.

Submitted 3 September 2025; accepted 17 March 2026

10.1126/science.aeb9953



An entropy-regulating molecular lock stabilizes formamidinium lead halide perovskite

Tianyin Miao, Sanwan Liu, Xia Lei, Yong Zhang, Wenpei Li, Qisen Zhou, Jianan Wang, Nikita A. Emelianov, Victoria V. Ozerova, Valeria S. Bolshakova, Wenqiang Wang, Zheng Zhou, Zhongjie Zhu, Lanlu Lu, Zhenhua Chen, Jingyuan Ma, Erxiang Xu, Luyao Wang, Yunfei Li, Zhengtian Tan, Shijie Zheng, Guilin Liu, Lianbo Guo, Jingbai Li, Yang Shen, Pavel A. Troshin, Sergey M. Aldoshin, Zonghao Liu, Nam-Gyu Park, and Wei Chen

Science **392** (6799), . DOI: 10.1126/science.aeb9953

Editor's summary

A molecular additive stabilizes the photoactive phase of formamidinium lead iodide (FAPbI₃) by increasing FA⁺ cation rotational entropy and suppressing the disordering of [PbI₆]⁴⁻ octahedra. Miao *et al.* showed that the dual-site flexible anchoring of 1-pyridin-3-ylmethyl-piperazine hydrochloride on the surface of FAPbI₃ films suppressed the formation of the inactive yellow phase at high temperature and humidity. A solar cell that incorporated a bismuth electrode maintained 93.0% of its power conversion efficiency of 26.8% after 1011 hours at 85°C. —Phil Szuromi

View the article online

<https://www.science.org/doi/10.1126/science.aeb9953>

Permissions

<https://www.science.org/help/reprints-and-permissions>

Use of this article is subject to the [Terms of service](#)

Science (ISSN 1095-9203) is published by the American Association for the Advancement of Science. 1200 New York Avenue NW, Washington, DC 20005. The title *Science* is a registered trademark of AAAS.

Copyright © 2026 The Authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original U.S. Government Works